

Dead or alive? Pitfall of survival analysis with TCGA datasets

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ABSTRACT

We often encounter situations in which data from the TCGA that have been analyzed in papers we read or reviewed cannot be reproduced, even when TCGA datasets are used, especially in survival analyses. Therefore, we attempted to confirm the data source for TCGA survival analysis and found that several websites used to analyze the survival data of TCGA datasets inappropriately handle the survival data, causing differences in statistical analyses. This causes the misinterpretation of results because figures of survival analysis results in several papers are sometimes exactly as generated by these sites, and the results depend on only the tools provided by these sites. We would like to make this situation widely known and raise the problem for scientific soundness.

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The analysis of clinical cohort studies is a very important and valuable method performed to confirm and validate the results of cancer research, especially in those from basic biological studies. However, cohort studies require considerable effort, cost, and time for one researcher. To address such a situation, The Cancer Genome Atlas (TCGA) project was launched in 2005 and has published huge amounts of datasets in various cancer tissues including comprehensive information on mutations, gene expression, copy number variation, and DNA methylation. Since these data are linked to detailed clinical information of each case including the tissue type, clinical stage, therapeutic regimen, and survival duration, we can analyze the relationship between genomic and clinical information in various cancers. TCGA project was summarized and completed as the Pan-Cancer Atlas.^{1–3} Although these data are publicly available and we can obtain valuable information by performing statistical analyses of these data, not all researchers are good at treating and analyzing such data adequately. Recently, several websites have provided tools that are relatively easy to use to analyze TCGA data, and many researchers have taken advantage of such sites for their studies. Additionally, the figures in a paper are sometimes exactly the same as those generated by these sites. However, we often encounter the situations in which TCGA data that have been analyzed in papers we read or reviewed cannot be reproduced, even if TCGA datasets are used, especially in survival analyses. Therefore, we attempted to confirm the data source used for TCGA survival analyses.

First, we compared the overall survival data between 11,315 cases from the GDC data portal (<https://portal.gdc.cancer.gov/>) in which all TCGA data have been deposited and 11,160 cases from the paper of Pan-Cancer Atlas⁴ which is recommended as a resource of survival data for analysis. We found differences in

days to death (death days) in 46 cases (increased except for one case) and in days to the last follow-up (follow days) in 56 cases (all increased) (Supplementary Table S1). Furthermore, the living status was changed in 24 cases. In 14 cases, the status was changed from dead to alive (Supplementary Table S2). If these differences in survival data are not adequately treated and processed, they may have a significant effect on survival analyses, resulting in misinterpretation.

We selected three widely used websites that provide tools that can be used to analyze survival data from TCGA, the Human Protein Atlas (<https://www.proteinatlas.org/>), KM plotter (<https://kmplot.com/analysis/>), and UCSC Xena (<http://xena.ucsc.edu/>). The provisional TCGA dataset in cBioPortal (<https://www.cbioportal.org/>) was also selected as a survey, although cBioPortal can adequately handle survival data from Pan-Cancer Atlas. We investigated 12 cases in whom the living status was changed from dead to alive with the information of days (Table 1). In most cases, day numbers were incorrectly imported. For example, the deaths of a total of six cases with urinary tract (BLCA), breast (BRCA), cervix (CESC), head and neck (HNSC), rectum (READ), and stomach (STAD) cancers were accompanied with follow days, not death days, in all three sites. In particular, the status change of one BRCA case (with 26 days) and one STAD case (with 21 days) may have significant effects on statistical analyses of survival because of the short-term deaths. In fact, the difference in survival data caused a significant change in the *p*-value (Supplementary figure). GEPIA (<http://gepia.cancer-pku.cn/>) also provides tools for survival analyses and many researchers use it. However, we were not able to verify the source for analysis because only graph figures, not text survival data from each case are included in the output of analysis. GEPIA provides not only overall survival (OS) but also disease-free

Table 1. Comparison of survival data at websites in TCGA cases which have different living statuses in GDC data portal and Pan-Cancer Atlas.

Case ID	Project	Tissue	GDC data portal			Pan-Cancer Atlas		Human Protein Atlas		KM plotter		UCSC Xena		cBioPortal TCGA provisional	
			Status	Death days	Follow days	Status	Days	Status	Days	Status	Days	Status	Days	Status	Days
TCGA-GD-A76B	BLCA	Urinary tract	Dead		224	Alive	224	Dead	224	Dead	224	Dead	224	Not included	
TCGA-E9-A245	BRCA	Breast	Dead		26	Alive	26	Dead	26	Dead	26	Dead	26	Not included	
TCGA-DS-A0VL	CESC	Cervix	Dead	1692	81	Alive	81	Alive	1692	Alive	1692	Dead	1692	Dead	1692
TCGA-06-0877	GBM	Glioblastoma	Dead		204	Alive	204	Not included		Not included		Dead	204	Dead	172
TCGA-H7-A6C4	HNSC	Head and neck	Dead		414	Alive	414	Dead	414	Dead	414	Dead	414	Not included	
TCGA-CL-4957	READ	Rectum	Dead		425	Alive	425	Dead	425	Dead	425	Dead	425	Not included	
TCGA-F5-6702	READ	Rectum	Dead	869	452	Alive	452	Alive	869	Alive	869	Dead	869	Dead	869
TCGA-DA-A1HW	SKCM	Melanoma	Dead	1096	820	Alive	820	Not included		Not included		Dead	1096	Dead	1096
TCGA-DA-A1IB	SKCM	Melanoma	Dead	1235	825	Alive	825	Not included		Not included		Dead	1235	Dead	1235
TCGA-BR-6707	STAD	Stomach	Dead	605	491	Alive	491	Alive	605	Alive	605	Dead	605	Dead	605
TCGA-BR-8380	STAD	Stomach	Dead		21	Alive	21	Dead	21	Dead	21	Dead	21	Not included	
TCGA-F1-6177	STAD	Stomach	Dead		0	Alive	200	Dead	200	Dead	200	Not included		Not included	

survival (DFS) data. Strangely, the case numbers of OS and DFS in GEPIA are exactly the same; nevertheless, TCGA provides DFS data on some cases. Thus, the analysis is “black box”.

It goes without saying that the survival analyses using TCGA datasets have great power for cancer research. However, it demonstrates its true value as far as the data is treated and analyzed adequately, as a matter of course. When the results of survival analyses generated by web tools are evaluated, even though the analysis is based on TCGA datasets, the underlying source of survival data should be confirmed (rather than easily accepting the results as is), although it is advisable that each researcher performs the survival analysis themselves with raw survival data provided by TCGA.

Disclosure of potential conflicts of interest

No potential conflicts of interest were disclosed.

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